



Eutrophication accelerated the phylogenetic evolution of *Daphnia sinensis* in Lake Chaohu based on palaeogenetic records

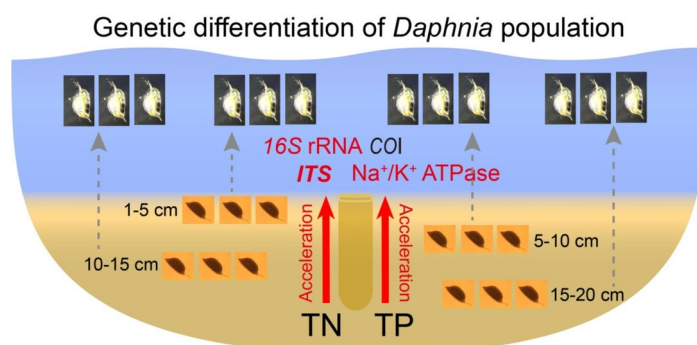
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Abstract

Increasing human activity and environmental changes have a complex impact on the evolutionary histories of most biomes. How environmental changes affect genetic evolution of wild populations has become a frequent research theme of biological ecology. Molecular techniques, in conjunction with palaeogenetics, facilitate understanding of the relationship between the genetic structure of wild populations and environmental change. In this study, genes encoding *16S* rRNA, cytochrome c oxidase subunit 1 (*COI*), internal transcribed spacer (*ITS*), and Na^+/K^+ ATPase were used as molecular markers to reconstruct changes in the temporal population genetic structure of *Daphnia sinensis* based on the analysis of resting eggs from the sediment layers in Lake Chaohu. The genetic variation of the *16S* rRNA, *COI*, *ITS*, and Na^+/K^+ ATPase genes of *D. sinensis* comprised 0–0.7%, 0–2.5%, 0–9.7%, and 0–15.4%, respectively, and was significantly affected by the total nitrogen and total phosphorus contents. The evolutionary molecular clock analysis predicted that sequences in different layers diverged over the time of 1.3–1.6 million years, which greatly exceeded the real age of the sediment (60 years). Phylogenetic analysis showed that genetic variation of the *16S* rRNA gene occurred the earliest, and the *ITS* gene had the highest genetic variation. Up to 32% of the Na^+/K^+ ATPase gene variation was explained by differences in the total phosphorus content. Moreover, we found evidence that high phosphorus levels may have triggered the directional selection of the Na^+/K^+ ATPase gene loci. Our results help to understand better the microevolution of cladoceran populations in eutrophic lakes.

Graphic abstract



Keywords *Daphnia sinensis* · Nitrogen · Phosphorus · Genetic diversity · Eutrophication · Lake Chaohu

Introduction

Over the last few centuries, the rapidly increasing anthropogenic changes have posed severe evolutionary challenges, although natural populations are adapting to the environmental changes (IPCC 2014; Pedersen et al. 2022). Increasing

human activity and environmental changes have profoundly affected the evolutionary histories of most biomes. Usually, aquatic environments are greatly affected (Schindler 1990; Chiaia-Hernandez et al. 2013), as nutrients such as nitrogen and phosphorus run off into rivers, lakes, and reservoirs. This alters biological populations and leaves behind distinctive signatures in the sediments (Schindler 1990), which could then be deciphered by palaeolimnological analyses (Rippey and Anderson 1996; Meng et al. 2023). The genetic analyses of egg banks suggested that these environmental changes can occur rapidly and affect the evolutionary trajectory of biological populations (Brede et al. 2009; Orsini et al. 2013a; Frisch et al. 2014). *Daphnia* resting egg banks preserved in lake sediments have been successfully used to reconstruct evolutionary processes (Decaestecker et al. 2007; Brede et al. 2009; Most et al. 2015). Insights into how populations respond and adapt to their environments can be obtained by integrating ecology and molecular biology approaches (Orsini et al. 2013b; O'Grady et al. 2022).

Lake Chaohu (117°18'–117°50' E, 31°25'–31°42' N) is a subtropical eutrophic shallow lake, with mean depth of 3 m (Deng et al. 2008; Wang et al. 2016). Lake Chaohu is one of five largest freshwater lakes in China and has a total surface area of 780 km². Because of severe eutrophication, cyanobacterial blooms have frequently occurred in Lake Chaohu since the 1970s (Tu et al. 1990; Deng et al. 2008; Wang 2010). In 1982–2015, although total nitrogen (TN) and total phosphorus (TP) concentrations showed obvious decreasing trends after 1995, they remained at high levels in Lake Chaohu (Zhang et al. 1997; Han 1998; Shang and Shang 2007; Wang 2007). Zhang (2005) found that TN and TP concentrations in sediments first increased, then decreased, and remained stable again with an increase in the sediment depth. There are many P-rich rocks on the north shore of the lake, which might be one of the important reasons for the relatively high P concentrations in the sediments of the lake (Tu et al. 1990). Eutrophication and cyanobacterial bloom outbreaks significantly affect seasonal dynamics and community succession of crustacean zooplankton, including *Daphnia* (Deng et al. 2008). During eutrophication, the genetic structure and evolutionary mechanisms within zooplankton populations in Lake Chaohu may also be strongly affected. The 16S rRNA, COI, and ITS genes have been widely used as molecular markers for the species identification and phylogeny of cladocerans in recent years (Taylor et al. 1998; Von et al. 1999; Dewaard et al. 2006; Richter et al. 2007; Elias-Gutierrez et al. 2008; Kotov et al. 2009; Xu et al. 2009; Wang et al. 2016). The Na⁺/K⁺ ATPase gene has also often been used in the studies of genetic variation of various organisms (Talsma et al. 2014; White et al. 2014; Arystarkhova 2016; Fan et al. 2016), but not in cladocerans. *Daphnia sinensis* (Gu et al. 2013) is widely distributed in the eutrophic waters of the middle and lower reaches of the

Yangtze River (Fig. 1) (Benzie 2005; Zhao 2005; Gu et al. 2013). Previous studies have mainly focused on the physiological and ecological aspects of *D. sinensis* biology (Xiang et al. 2010; Huang et al. 2015; Zhang et al. 2015; Jiang et al. 2016), however, understanding of the genetic structure and evolutionary mechanisms shaping *D. sinensis* populations under environmental changes is insufficient, especially for wild populations in the natural state.

In this study, the genetic variation and phylogeny of *D. sinensis* hatched from resting eggs in the sediments of Lake Chaohu were analysed based on the sequences of the 16S rRNA, COI, ITS, and Na⁺/K⁺ ATPase genes. We expected to be able to test the following hypothesis: 1) the lake eutrophication would accelerate the genetic difference of *D. sinensis*; 2) there may be significant difference in the responses of four genes to lake eutrophication; 3) the change of phosphorus content would promote the genetic mutations in Na⁺/K⁺ ATPase gene. Our study will help further to understand the

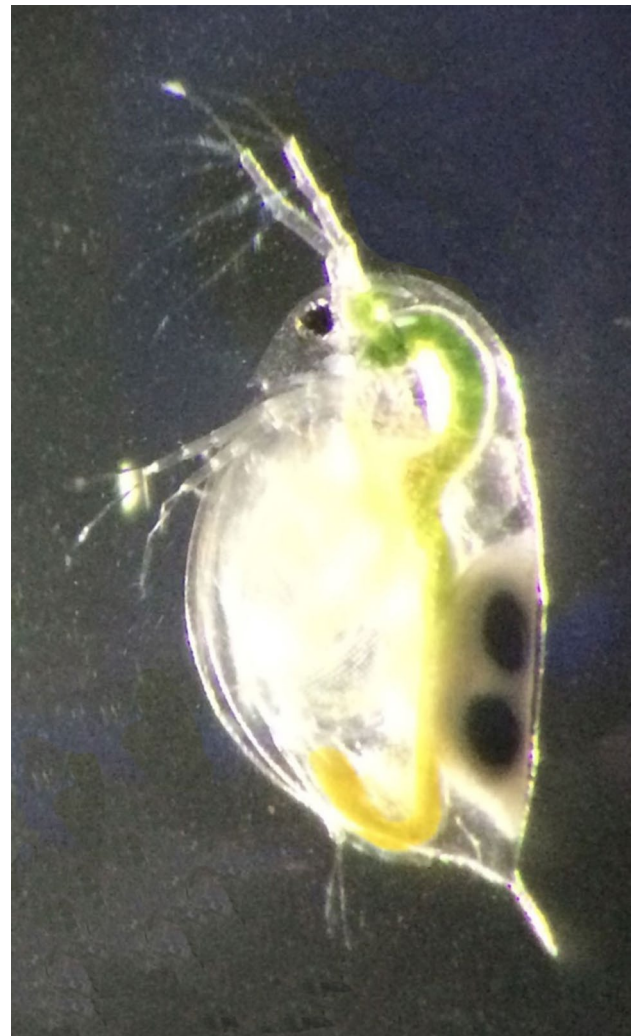


Fig. 1 Photograph of an adult *D. sinensis* female and its resting eggs

microevolution occurrence of *Daphnia* species in eutrophic lakes.

Materials and methods

Sampling site and sediment cores

The sampling site was located in the central zone of Lake Chaohu (31°33′28.74″ N, 117°32′11.41″ E) (Online resource, Fig. S1). Four sediment cores were collected using a columnar gravity corer 4.2 cm in diameter (Nanjing Institute of Geography & Limnology, Chinese Academy of Science) in August 2014 (Zhang et al. 2020). The physical and chemical characteristics of the lake water during the sampling period are shown in Table 1. Each sediment core was cut into 1-cm thick sections (samples) and stored in a refrigerator at 4 °C.

The samples of one sediment core were dried and weighed using an electronic scale. The samples were dried in an oven under the condition of 105 ± 5 °C. The drying time is set according to the quantity and humidity of the sample, which generally takes 16 to 24 h. Cooling and weighing: The dried sample is cooled to room temperature in the dryer. Dry samples were crushed and sieved through a 0.154-mm mesh. 2–5 g powdered sample was weighed, and the age of the sediment depth was determined by a high purity germanium gamma spectrometer (Ametek-Amt Ortec C., China) at the State Key Laboratory of Lake Science and Environment, Nanjing Institute of Geography and Limnology, Chinese Academy of Sciences. According to the attenuation law of radioactive isotope, the radioactive dates were calculated using the constant rate of supply (CRS) ^{210}Pb dating model (Chen et al. 2014). The ground samples were filtered using a 0.074-mm mesh, and then, 50 mg of each 1-cm thick filtered sample was weighed to measure TN and TP contents by alkaline potassium persulfate digestion UV spectrophotometry (Qian et al. 1990).

The remaining three sediment cores were used to collect the ephippia, i.e., resting eggs of *D. sinensis* (Fig. 1). Sediment samples (thick) were rinsed with aerated tap water and filtered through a plankton net (64 µm). The residual sediment was observed under a microscope (Olympus, Japan). *D. sinensis* ephippia and hatched individuals were identified according to the methods described by Benzie (2005).

Table 1 Physico-chemical parameters measured in Lake Chaohu during the sampling period

Sampling site	pH	Temperature (°C)	Dissolved oxygen (mg L ⁻¹)	Depth (m)	Transparency (m)
Central zone	8.65	26.2	7.23	3.5	0.45

Total DNA extraction

The resting eggs of *D. sinensis* from different sediment depths were incubated and hatched in an intelligent light incubator (Ningbo Saifu, China) at 25 °C. Each neonate was cultured at 25 °C, with the 12 h light:12 h darkness illumination cycle. *Tetrademus obliquus* was used as food, and *D. sinensis* were kept in aerated tap water for 48 h (Wang et al. 2016).

The 102 young *D. sinensis* (Fig. 1) hatched from resting eggs were individually cultured to become adults at 25 °C. Genomic DNA of each adult *D. sinensis* was extracted using a TIANamp Micro DNA Kit (Wang et al. 2016). The integrity, concentration, and quality of the DNA were checked using a Nanodrop 2000 instrument (Thermo Fisher Scientific, USA). The concentration of DNA was 85–120 ng/µL, and the A260/A280 ratio was 2.0–2.3.

PCR amplification and sequence length measurements

Mitochondrial 16S rRNA was amplified using the L2510 and H3059 primers (Bouchon et al. 1994; Xu et al. 2014). The COI gene was amplified with the COIF and COIR primers (Xu et al. 2014; Wang et al. 2016). The ITS gene was amplified with the ITSF (5.8BF) (5′-CCCTGAACG GTGGATCACTA-3′) and ITSr (5.8BR) (5′-TTAGAA GGAGTTTACCTCCCGCTTAGG-3′) primers (Crease 1999; Taylor et al. 2005). The Na⁺/K⁺ ATPase gene was amplified with the P₁F (5′-TACATGGAGCTTGGTGA ATG-3′) and P₁R (5′-GCAAAATCATGTCTGCAGCTT-3′) primers. The design of the Na⁺/K⁺ ATPase gene primers was done based on the *D. sinensis* transcriptome data (Zhang et al. 2016).

The 25 µL PCR mixture contained 1.0 µL of the genomic DNA, 2.5 µL of 10×LA-Taq Buffer II, 0.25 µL of DNA polymerase TaKaRa-LA-Taq (5 U/µL) (Clontech, USA), 4.0 µL of dNTPs (2.5 mM) (Sangon Biotech, Shanghai), 1.0 µL of each primer (10 mM) (Sangon Biotech, Shanghai), 0.5 µL of Mg²⁺ (25 mM), and 14.75 µL of double-distilled H₂O.

The conditions for 16S rRNA and COI gene amplification were described by Wang et al. (2016). For the ITS gene, an initial denaturing step of 2 min at 94 °C was followed by 35 cycles of [1 min at 94 °C, 45 s at 53 °C, and 2 min at 72 °C] and finished with the final extension step at 72 °C for 10 min. For the Na⁺/K⁺ ATPase gene, an initial denaturing step of 3 min at 94 °C was followed by 35 cycles of [30 s at 94 °C, 30 s at 58 °C, and 1 min at 72 °C], and finished with the final extension step at 72 °C for 10 min. PCR products were purified using a TIANGEN Biotech Kit (Tiangen, Beijing) and their lengths were confirmed by electrophoresis in a 1% agarose gel.

Alignment and phylogenetic analysis

The PCR products were analysed by gel electrophoresis and sequenced (GenScript, Nanjing, China). The *16S* rRNA (valid bases 508–523), *COI* (valid bases 489–531), *ITS* (valid bases 1497–1524), and Na^+/K^+ ATPase (valid bases 734–736) genes were detected. The bidirectional nucleotide sequences were aligned and proofread using DNASTar (DNASTAR Inc., Madison, WI, USA) to remove unreliable bases based on the SeqMan peak. The homologous sequences used as outgroup for the analysis were downloaded from GenBank (Table 2).

Multiple sequence alignments were performed using ClustalX (Larking et al. 2007). The variable sites of *D. sinensis* sequences were analysed using DNAspV5 (Librado and Rozas 2009). The conversion/transversion ratios and genetic distances of *D. sinensis* at different sediment depths were calculated using MEGA 6.0 (Tamura et al. 2013). Genetic distances were calculated using the Kimura two-parameter model with 1,000 bootstrap replicates. PhyML 3.0 (Guindon et al. 2010) was used for the maximum likelihood (ML) analysis utilising the GTR + G + I evolutionary model (Modeltest version 3.7, Nylander, 2008). Phylogenetic trees were constructed using Bayesian inference in MrBayes 3.1.2 (Ronquist et al. 2003). The first 25% of the trees were discarded as burn-in, and a 50% majority-rule consensus tree was constructed from the remaining Bayesian trees based on 10,000,000 generations (Wang et al. 2016). *Daphnia obtusa* (Kurz, 1874) (FJ427466) (Adamowicz et al. 2009), *D. pulex* (Leydig, 1860) (KF99371) (Xu et al. 2014), *D. galeata* (G. O. Sars, 1863) (AB642075) (Ishida et al. 2011), and *D. pulex* (NC_000844) (Crease 1999) were used to construct the phylogenetic trees as outgroups of the *16S* rRNA, *COI*,

ITS, and Na^+/K^+ ATPase genes of *D. sinensis*. Based on calibrations for other crustacean groups, divergence times were estimated using homologous *16S* rRNA and *COI* gene fragments for comparison.

Groups in the sediments

The resting eggs of *D. sinensis* in the 20 cm sediment depth of Lake Chaohu were divided into four groups at 5-cm intervals to avoid mixed interference referencing the age of the sediment depth (i.e., A group: 1–5 cm ~ age of 2008–2014, B group: 5–10 cm ~ age of 1996–2008, C group: 10–15 cm ~ age of 1966–1996, and D group: 15–20 cm ~ age of 1954–1966). Analysis of molecular variance (AMOVA) was conducted to analyse the diversity of nucleic acid sequences of *D. sinensis* from different sediment layers of Lake Chaohu using Arlequin 3.5 (Excoffier and Lischer 2010), in which all individuals hatched from *D. sinensis* resting eggs obtained from the same 5-cm interval were regarded as belonging to the same group.

Relationships between nutrient factors and genetic variation of *D. sinensis*

Redundancy analysis (RDA) and variation partitioning analysis were employed to test the relationships between genetic variation of *D. sinensis* and TN/TP contents in the sediments using CANOCO for Windows ver. 4.5 (ter Braak and Smilauer 2002). In addition, one-way ANOVA was used to assess the relative impact of TN and TP on genetic variation of *D. sinensis*, using the “vegan” package (version 2.6-8) in R software (Oksanen et al. 2024).

Table 2 Codes of *Daphnia* species used in this study

Species	Sediment depth	Collection catalog codes			
		<i>16S</i> rRNA	<i>COI</i>	<i>ITS</i>	Na^+/K^+ ATPase
<i>D. sinensis</i>	A (1–5 cm)	A1-A21	A1-A21	A1-A21	A1-A21
<i>D. sinensis</i>	B (6–10 cm)	B1-B42	B1-B42	B1-B42	B1-B42
<i>D. sinensis</i>	C (11–15 cm)	C1-C28	C1-C28	C1-C28	C1-C28
<i>D. sinensis</i>	D (16–20 cm)	D1-D11	D1-D11	D1-D11	D1-D11
<i>D. obtusa</i>		FJ427466			
<i>D. pulex</i>			KF99371		
<i>D. galeata</i>				AB642075	
<i>D. pulex</i>					NC_000844

16S rRNA: GenBank accession numbers: MK922363-MK922464 (A1-A21, B1-B42, C1-C28, D1-D11); *COI*: GenBank accession numbers: MK941503-MK941604 (A1-A21, B1-B42, C1-C28, D1-D11); *ITS*: GenBank accession numbers: MK940963-MK941064 (A1-A21, B1-B42, C1-C28, D1-D11); Na^+/K^+ ATPase: GenBank accession numbers: MK952040-MK952141 (A1-A21, B1-B42, C1-C28, D1-D11)

Results

Age of Lake Chaohu sediments

^{210}Pb dating showed that the 1–20 cm sediment depth of Lake Chaohu represented ~60 years, which is consistent with the estimates of Zhang (2005) (Online resource, Fig. S2). TN and TP in the sediments were initially high, then low, and then remained stable with the increase in the sediment depth (Online resource, Fig. S3).

Molecular variation inferred from the sequences of the 16S rRNA, COI, ITS, and Na⁺/K⁺ ATPase genes in *D. sinensis* resting eggs

The 102 sequences of the *D. sinensis* 16S rRNA gene contained 507 identifiable base sites, including 498 conserved, nine variable, four single, and five parsimony-informative sites. The GenBank accession numbers for these sequences are MK922363–MK922464 (A1–A21, B1–B42, C1–C28, and D1–D11; Table 2). The average A, T/U, C, and G contents were 34.4%, 34.4%, 18.2%, and 13.0%, respectively. The A + T content (68.8%) was significantly higher than the G + C content (31.2%). The overall transition/transversion ratio was 1.0.

The 102 sequences of the COI gene contained 486 identifiable base sites, including 465 conserved, 21 variable, 2 single, and 19 parsimony-informative sites. The GenBank accession numbers for these sequences are MK941503–MK941604 (A1–A21, B1–B42, C1–C28, and D1–D11; Table 2). The average A, T/U, C, and G contents were 23.8%, 40.0%, 14.4%, and 21.8%, respectively. The A + T content (63.8%) was significantly higher than the G + C content (36.2%), with an overall transition/transversion ratio of 2.0.

The 102 sequences of the ITS gene contained 1,468 identifiable sites, including 1,116 conserved, 352 variable, 134 single, and 218 parsimony-informative sites. The GenBank accession numbers for these sequences are MK940963–MK941064 (A1–A21, B1–B42, C1–C28, and D1–D11; Table 2). The average A, T/U, C, and G contents were 23.7%, 22.6%, 25.0%, and 28.8%, respectively. The A + T content (46.3%) was lower than the G + C content (53.7%), with an overall transition/transversion ratio of 0.5.

The 102 sequences of the Na⁺/K⁺ ATPase gene contained 728 identifiable sites, including 558 conserved, 170 variable, three single, and 167 parsimony-informative sites. The GenBank accession numbers for these sequences are MK952040–MK952141 (A1–A21, B1–B42, C1–C28, and D1–D11; Table 2). The average A, T/U, C, and G contents were 23.4, 29.3, 24.8, and 22.5%, respectively. The A + T content (51.7%) was slightly higher than that of G + C (47.3%), with an overall transition/transversion ratio of 0.67.

The sequence composition of the 16S rRNA gene (A + T content: 68.8%, G + C content: 31.2%, transition/transversion ratio: 1.0) was similar to that of the COI gene (A + T content: 63.8%, G + C content: 36.2%, transition/transversion ratio: 2.0). The sequence composition of the ITS gene (A + T content: 46.3%; G + C content: 53.7%; transition/transversion ratio: 0.5) was closer to that of the Na⁺/K⁺ ATPase gene (A + T content: 51.7%; G + C content: 47.3%; transition/transversion ratio: 0.67). ITS and Na⁺/K⁺ ATPase genes were more variable than 16S rRNA and COI genes.

In addition, AMOVA showed significant genetic variation of the 16S rRNA, COI, and ITS genes ($p < 0.001$) among and within populations of *D. sinensis* (Table 3). Moreover, there were significant genetic differences ($p < 0.001$) between the 1–15 cm and 15–20 cm sediment layers for the 16S rRNA gene, between the 1–5 cm and 5–20 cm sediment layers for the COI gene, and between the 1–10 cm and 10–20 cm sediment layers for the ITS gene (Table 4).

Table 3 Analysis of molecular variance between the populations and within the population based on the 16S rRNA, COI, and ITS gene sequences in *D. sinensis* resting eggs from different sediment layers in Lake Chaohu

	Source of variation	Degrees of freedom	Sum of squares	Variance components	Variation (%)	<i>p</i> -value	FST
16S rRNA	Among population	3	16.630	0.205 Va	24.18	0.000	0.242
	Within population	98	63.165	0.645 Vb	75.82		
	Total	101	79.794	0.850			
COI gene	Among population	3	548.667	7.411 Va	54.32	0.000	0.543
	Within population	98	610.764	6.232 Vb	45.68		
	Total	101	1159.431	13.643			
ITS gene	Among population	3	145.684	1.308 Va	7.01	0.000	0.070
	Within population	98	1701.894	17.366 Vb	92.99		
	Total	101	1847.578	18.674			

Va and Vb is the coefficient of variation in AMOVA. A population stands for a group at 5 cm interval of sediment depths

Table 4 Analysis of molecular variance between different groups based on the *16S* rRNA, *COI*, and *ITS* gene sequences in *D. sinensis* resting eggs from different sediment layers in Lake Chaohu

	Source of variation	Degrees of freedom	Sum of squares	Variance components	Variation (%)	FSC/FST	<i>p</i> -value
<i>16S</i> rRNA	Among groups	1	14.062	0.669 Va	50.08	0.033/0.517	0.000/0.014
	Among populations within groups	2	2.568	0.022 Vb	1.65		
	Within populations	98	63.165	0.645 Vc	48.27		
	Total	101	79.794	1.335			
<i>COI</i> gene	Among groups	1	316.086	6.068 Va	35.94	0.424/0.631	0.000/0.000
	Among populations within groups	2	232.582	4.581 Vb	27.14		
	Within populations	98	610.764	6.232 Vc	36.96		
	Total	101	1159.431	16.881			
<i>ITS</i> gene	Among groups	1	64.195	0.357Va	1.9	0.058/0.076	0.000/0.000
	Among populations within groups	2	81.489	1.067Vb	5.68		
	Within populations	98	1701.894	17.366Vc	92.41		
	Total	101	1847.578	18.791			

Populations were evaluated as two groups (*16S* rRNA: D population vs. A, B and C populations; *COI*: A population vs. B, C and D populations; *ITS*: A and B populations vs. C and D populations)

Phyletic evolution of the *16S* rRNA *COI*, *ITS*, and Na^+/K^+ ATPase genes in *D. sinensis* resting eggs in different sediment layers

The genetic distances of the *16S* rRNA gene ranged from 0 to 0.007, based on the Kimura 2-parameter distance. Phylogenetic trees showed highly congruent tree topologies. The main divergences in the ML tree were in accordance with those in the MrBayes and NJ trees (Fig. 2). In the *16S* rRNA phylogenetic tree, *D. sinensis* from different sediment depths evolved into two branches. One branch included *D. sinensis* from the 15–20 cm sediment layer (group D), and the other branch included *D. sinensis* from the 1–15 cm sediment layers (groups A, B, and C). In the *16S* rRNA from the 15–20 cm sediment layer, base No. 54 was T, whereas in other sediment layers, it was base A. In the sequences from the 15–20 cm sediment layer, base No. 58 was G, whereas there was no base in that position in the other sediment layers. These results suggest an obvious divergence of some *D. sinensis* *16S* rRNA loci between the sequences isolated at a depth of 1–15 cm compared to those isolated at the 15–20 cm sediment layer in Lake Chaohu.

The genetic distances of the *COI* gene ranged from 0 to 0.025. In the *COI* phylogenetic trees (Fig. 3), group B (5–10 cm) showed greater genetic diversity than the other groups. However, 52% of the individuals in group A (1–5 cm) showed obvious divergence from those in groups C and D (10–20 cm). With regards to differences in sequences, base No. 7 in the 1–5 cm sediment layers was T, whereas it was C in the 10–15 cm sediment layers. Base No. 81 in the 1–5 cm sediment layers was T, whereas it was C in the 5–20 cm sediment layers.

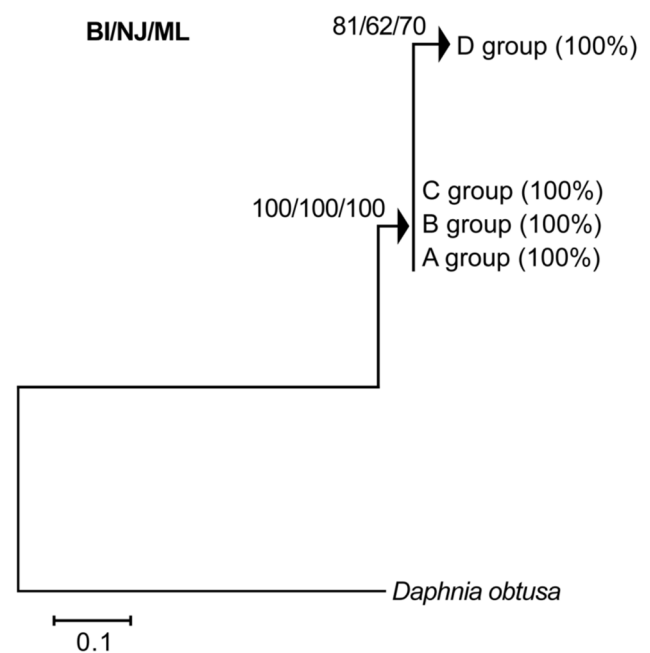


Fig. 2 *Daphnia sinensis* phylogeny inferred from the *16S* rRNA sequences as a consensus tree formed from the trees constructed using the maximum likelihood (ML), neighbour-joining (NJ), and Bayesian inference (BI) methods. The number in parenthesis represents the percent of sequences in total sequences of each group

The range of genetic distances of the *ITS* gene was 0–9.7%. The main divergences of the *ITS* ML phylogenetic tree were in accordance with those of the MrBayes and NJ trees (Fig. 4). The phylogenetic tree of the *ITS* gene didn't show obvious regular variations; however, some base sites were characterised by a particular variability pattern. In the

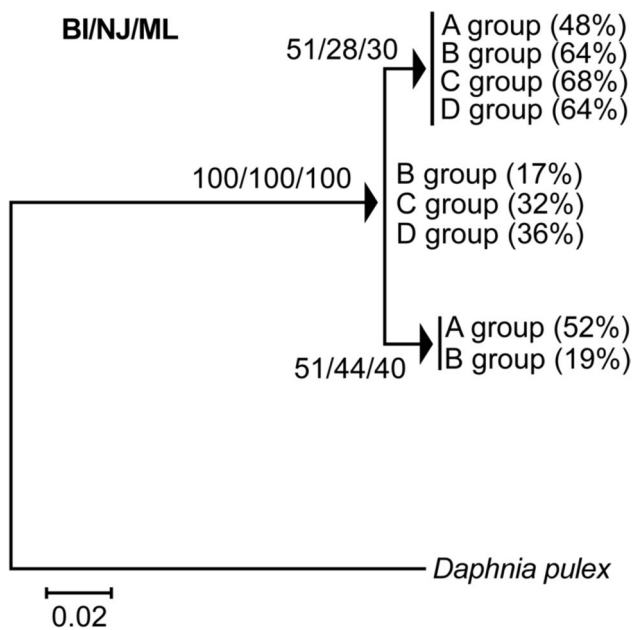


Fig. 3 *Daphnia sinensis* phylogeny inferred from *COI* gene sequences as a consensus tree formed from the trees constructed using the maximum likelihood (ML), neighbour-joining (NJ), and Bayesian inference (BI) methods. The number in parenthesis represents the percent of sequences in total sequences of each group

ITS gene sequences of *D. sinensis*, bases 205–925 were a large variable region. Base No. 135 in the 1–10 cm sediment layers was C, whereas it was T in the 10–20 cm sediment

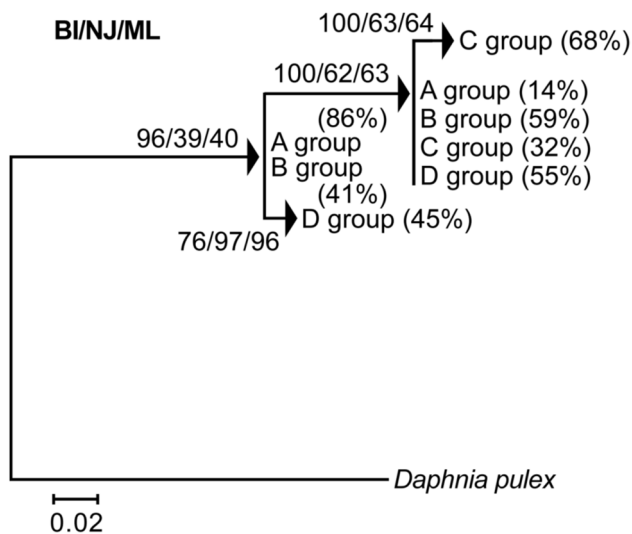


Fig. 4 *Daphnia sinensis* phylogeny inferred from the *ITS* gene sequences as a consensus tree formed from the trees constructed using the maximum likelihood (ML), neighbour-joining (NJ), and Bayesian inference (BI) methods. The number in parenthesis represents the percent of sequences in total sequences of each group

layers. In addition, base No. 146 in the 1–10 cm sediment layers was T, whereas it was C in the 10–20 cm sediment layers.

The Na^+/K^+ ATPase gene genetic distances ranged from 0% to 15.4% based on the Kimura 2-parameter distance. The main divergences in the Na^+/K^+ ATPase gene ML phylogenetic tree were in accordance with those in the MrBayes and NJ trees (Fig. 5). Regular variations of base sites in the Na^+/K^+ ATPase loci of *D. sinensis* were weaker than those in the other three genes. However, divergences at some base sites of the Na^+/K^+ ATPase gene were also obvious between the 5–15 cm sediment layers on the one hand and the 1–5 cm and 15–20 cm sediment layers on the other hand. Base No. 135 from the 5–15 cm sediment layers was C, whereas it was T in both the 1–5 cm and 15–20 cm sediment layers. Base No. 162 from the 5–15 cm sediment layers was T, whereas it was A in both the 1–5 cm and 15–20 cm sediment layers.

Influence of TN and TP contents on the genetic variation in *D. sinensis* resting eggs

RDA showed that *16S* rRNA, *COI*, *ITS*, and Na^+/K^+ ATPase gene sequences of *D. sinensis* were significantly affected by TN and TP contents in different sediment layers (Figs. 6, 7).

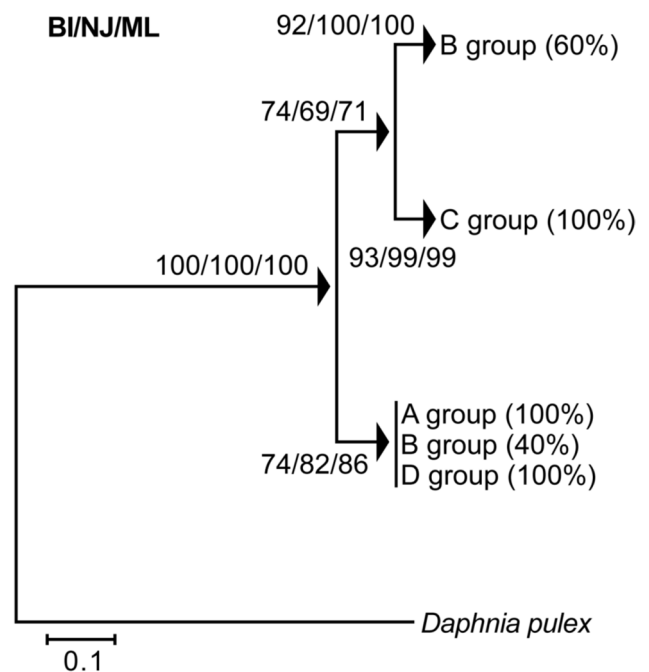
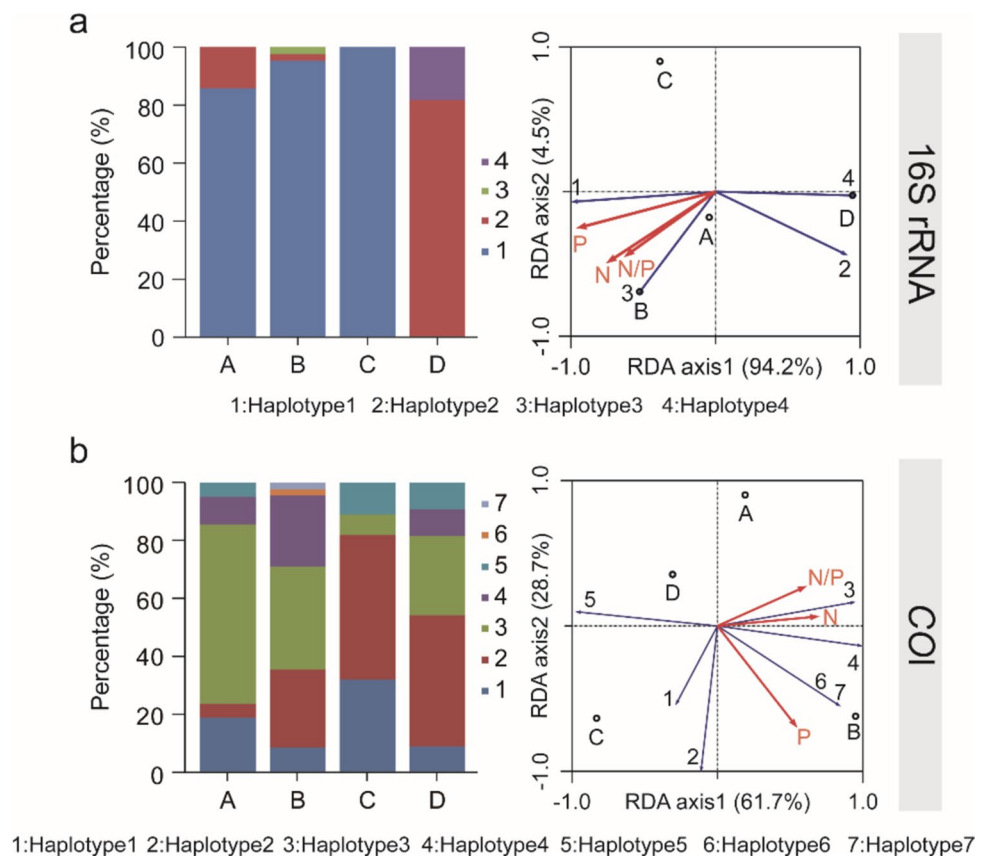


Fig. 5 *Daphnia sinensis* phylogeny inferred from the Na^+/K^+ ATPase gene sequences as a consensus tree formed from the trees constructed using the maximum likelihood (ML), neighbour-joining (NJ), and Bayesian inference (BI) methods. The number in parenthesis represents the percent of sequences in total sequences of each group

Fig. 6 Relationships between historic population genetic structure of the *16S* rRNA (a) and *COI* (b) genes of *D. sinensis* and environmental variables in different sediment layers from Lake Chaohu. N/P: Ratio of the total nitrogen to total phosphorus content, TN: total nitrogen content, TP: total phosphorus content. Different sediment layers: A: 1–5 cm, B: 5–10 cm, C: 10–15 cm, D: 15–20 cm. Factor significance level $p < 0.01$



One-way ANOVA showed the influence of TN and TP contents on the genetic variation of the four genes (Table 5). Moreover, there was a significant correlation between TP content and genetic variation of the Na^+/K^+ ATPase gene ($r=0.845$, $p < 0.01$). TP explained 32% of the genetic variation in the Na^+/K^+ ATPase gene.

Discussion

The A + T content in the *16S* rRNA and *COI* gene sequences of *D. sinensis* from different sediment layers of Lake Chaohu (68.8% and 63.8%, respectively) was significantly higher than that of G + C (31.2% and 36.2%, respectively), which was consistent with the generally higher A + T content among invertebrates (Anderson 2000; Gao et al. 2000). However, for the *ITS* gene, the A + T content (46.3%) was lower than the G + C content (53.7%). The overall transition/transversion ratios of the *ITS* (0.5) and Na^+/K^+ ATPase genes (0.65) were significantly lower than those of the *16S* rRNA (1) and *COI* genes (2). These results reflect the structural specificity of the *ITS* and Na^+/K^+ ATPase gene sequences in *D. sinensis* (Taylor et al. 2005; White et al. 2014; Arystarkhova 2016). There were small differences between different ages in the base percentage and overall transition/transversion ratios of *16S* rRNA, *COI*, *ITS*, and Na^+/K^+ ATPase genes.

The variability (1.8%) and genetic distance (0.7%) of *D. sinensis* *16S* rRNA sequences from different sediment layers of Lake Chaohu were consistent with the highly conserved characteristics of this gene, in line with reports by Taylor et al. (1998) and Dai et al. (2010). For the *D. sinensis* *COI* gene sequences, variability and genetic distance estimates comprised 4.3% and 2.5%, respectively. Compared with the gene sequences of *16S* rRNA and *COI*, the *ITS* and Na^+/K^+ ATPase genes had higher variability (24% and 23%, respectively) and genetic distances (9.7% and 15.4%, respectively), which correlated with previously noted highly variable features of these genes (Crease 1999; Taylor et al. 2005; Kotov and Taylor 2010; Ishida et al. 2011; Didehdar et al. 2015).

Owing to the development of industry and agriculture accompanied by elevated levels of human activity, Lake Chaohu has suffered from eutrophication since the 1970s (Tu et al. 1990; Zhang and Kong 2015). During eutrophication, the genetic structure and evolutionary mechanisms regulating zooplankton populations in Lake Chaohu were likely strongly affected. In this study, RDA indicated that TN and TP had significant effects on the genetic diversity of *D. sinensis* resting eggs based on the analysis of *16S* rRNA, *COI*, *ITS*, and Na^+/K^+ ATPase gene sequences ($p < 0.01$) from different sediment layers of Lake Chaohu. Variations in TN and TP collectively explained > 50% of the genetic diversity of *D. sinensis* resting eggs inferred from *16S* rRNA,

Fig. 7 Relationships between historic population genetic structure of the Na^+/K^+ ATPase (a) and *ITS* (b) genes of *D. sinensis* and environmental variables in different sediment layers from Lake Chaohu. N/P: Ratio of the total nitrogen and total phosphorus content, TN: total nitrogen content, TP: total phosphorus content. Different sediment layers: A: 1–5 cm, B: 5–10 cm, C: 10–15 cm, D: 15–20 cm. Factor significance level $p < 0.01$

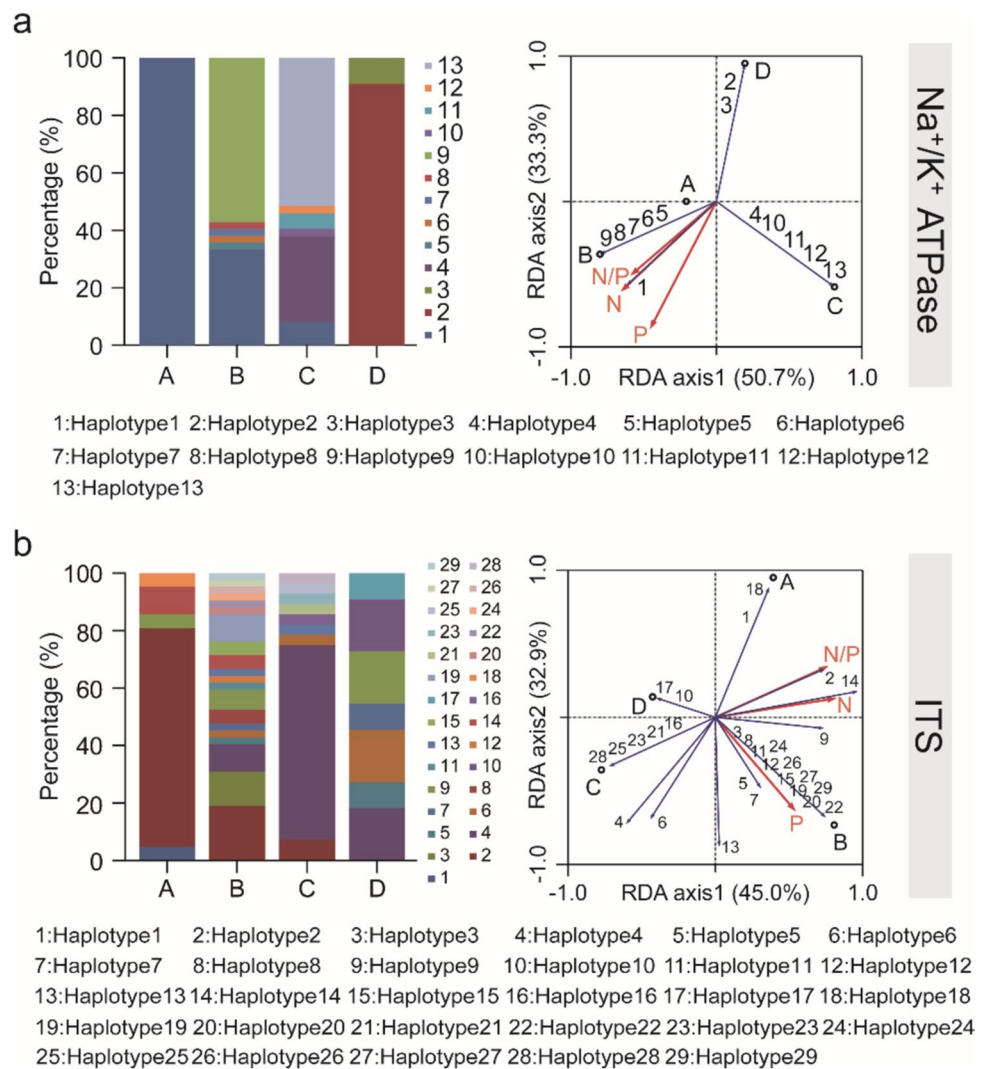


Table 5 One-way analysis of variance results showing the effects of total nitrogen and total phosphorus contents on the variation of the *16S* rRNA, *COI*, *ITS*, and Na^+/K^+ ATPase genes in *D. sinensis* resting eggs from different sediment layers in Lake Chaohu

Gene name	TN R-squares	TP R-squares
<i>16S</i> rRNA gene	0.387	0.359
<i>COI</i> gene	0.248	0.247
<i>ITS</i> gene	0.458	0.204
Na^+/K^+ ATPase gene	0.228	0.316

Bold fonts represent significance at $p < 0.05$

COI, *ITS*, and Na^+/K^+ ATPase gene sequences. One-way ANOVA analysis showed that TN differences explained 39% and 46% of genetic variation in the *16S* rRNA and *ITS* genes. In turn, TP differences explained 36% and 32% of genetic variation in the *COI* and Na^+/K^+ ATPase genes. ^{210}Pb dating showed that the 1–20 cm sediment depth of Lake Chaohu

represents approximately 60 years, which is similar to the estimates reported by Zhang (2005). However, according to the evolutionary molecular clock method (Taylor et al. 1998; Vergilino et al. 2011), genetic distances of the *16S* rRNA (0.7%) and *COI* genes (2.5%) in *D. sinensis* resting eggs found at different depths of the 1–20 cm sediment layers was about 1.3–1.6 million years. Therefore, we concluded that the process of eutrophication significantly accelerated genetic variation in *D. sinensis* population in Lake Chaohu, which is similar to the findings of Frisch et al. (2014). Lake eutrophication plays a key role in the genetic variation of natural populations of *Daphnia* species.

Although eutrophication significantly accelerated the evolutionary speed of genetic variation in *D. sinensis*, the responses of the *16S* rRNA, *COI*, *ITS*, and Na^+/K^+ ATPase genes to eutrophication in Lake Chaohu varied. Genetic variation of the *16S* rRNA gene occurred earlier than that of the *COI* gene during eutrophication. The TN and TP concentrations had a greater effect on the *16S* rRNA gene, explaining

39% and 36% of its variation, respectively. In the case of the *ITS* gene, TN had a greater effect than TP, accounting for 46% of the *ITS* gene variation. Variation of the *ITS* gene was particularly high during eutrophication, with 29 haplotypes observed. The Na^+/K^+ ATPase gene had 13 gene haplotypes, and TP had the greatest influence on this gene, accounting for 32% of its variation. Greater and more regular base variations in the Na^+/K^+ ATPase gene sequences of *D. sinensis* resting eggs from the 5–15 cm sediment layers were observed compared to those in the 1–5 cm and 15–20 cm sediment layers, which was clearly reflected in the phylogenetic trees (Fig. 5). ^{210}Pb dating showed that the 5–15 cm sediment layers of Lake Chaohu formed in 1976–1998, when TP was higher than at other sediment depths. Similarly, higher phosphorus concentrations in Lake Chaohu water were observed during 1976–1989 (Zhang et al. 1997; Han 1998; Shang and Shang 2007; Wang 2007). Moreover, previous studies indicated that the population size of *Daphnia* increased gradually during this period (Deng et al. 2008). The relationship between the TP concentration and genetic variation of the Na^+/K^+ -ATPase gene sequence was also significant ($r=0.845$, $p<0.01$). The TP concentration explained 32% of the genetic changes in the Na^+/K^+ ATPase gene. These results suggest that long-term changes in TP concentration in the lake directly or indirectly impacted genetic variation of the *D. sinensis* Na^+/K^+ ATPase gene under natural selection, which is in accordance with the results of Orsini et al. (2013b). In addition, differences in bases in some Na^+/K^+ ATPase gene loci were obvious between *D. sinensis* resting eggs in the 5–15 cm layer compared to those in the 1–5 cm and 15–20 cm sediment layers in Lake Chaohu. For example, base No. 135 site in the 5–15 cm sediment layer was C, whereas it was always T in the eggs from both the 1–5 cm and 15–20 cm sediment layers. In addition, base No. 162 in the 5–15 cm sediment layer was T, whereas it was A in resting eggs from both the 1–5 cm and 15–20 cm sediment layers. Moreover, the TP content in the 5–15 cm sediment layer was significantly higher than those in the 1–5 cm and 15–20 cm sediment layers. Therefore, we deduced that high TP content may trigger the directional selection of bases in the Na^+/K^+ ATPase gene. This study has certain limitations due to research methodology. Only four molecular markers were utilized, which might not comprehensively represent the entire molecular profile of the subject under investigation. The limited number of markers could lead to an incomplete understanding of the genetic complexity and variation, potentially overlooking significant genetic factors.

In general, we conclude that the microevolution of *D. sinensis* in Lake Chaohu is a complex process and may be significantly affected by lake eutrophication. For future research, it is essential to expand the scope of molecular markers and add more environmental factors. Combine a

broader range of markers with the influence of multiple environmental factors would provide a more detailed and accurate picture of the genetic landscape. This would help to overcome the current limitations and enhance the overall understanding and significance of the research.

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Data availability The original data are available from The GenBank database (NCBI database, the link: <https://www.ncbi.nlm.nih.gov/>). The GenBank accession numbers of the *16S* rRNA, *COI*, *ITS*, and Na^+/K^+ ATPase sequences are, respectively, MK922363–MK922464, MK941503–MK941604, MK940963–MK941064, and MK952040–MK952141.

Declarations

Conflict of interest The authors declare that this study was conducted in the absence of any commercial or financial relationships that could be construed as potential conflicts of interest.

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